

Kevin Jaewoo Lee

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PROJECT

Ticket Booking Web Service

- Developed a highly consistent ticket-booking backend using DynamoDB with Node.js in a serverless environment with Lambda and API Gateway, implementing TTL-based seat locks to handle real-time reservations reliably.
- Optimized search engine by indexing event name and location queries using Amazon OpenSearch. Designed custom query and keyword analyzers (edge-ngram, autocomplete, synonyms) to detect full-text and structured filtering for ticket-booking data.
- Implemented presigned URLs to securely upload event posters and seat maps directly from clients to Amazon S3, reducing average TTFB from 600.7 ms to 30.2 ms, achieving a 94.5% improvement.

University Course Selection Application

- Developed a full-stack web application using Express.js and React.js, integrating Firebase for user registration and authentication.
- Integrated with MongoDB, enabling management of user data, course selection, and chatroom services, allowing users to register for courses and communicate within the corresponding course.
- Designed a real-time chat service using WebSockets to support instant message broadcasting and tracking user presence, enabling seamless communication between users enrolled in the same course.

WORK EXPERIENCE

Software Developer

03/2025 - 08/2025

Grow Your Business

Norwalk, Connecticut

- Developed a real-time chat service powered by Firebase Firestore and React, implementing thread creation and snapshot listeners for seamless updates. Refactored chat front-end to state-driven rendering model, restructuring the component to ensure stable and smooth multi-user interactions.
- Refactored a single-assistant model into multiple role-specialized agents using OpenAI API with custom fine-tuning to reflect company tone, goal, and product knowledge, improving response accuracy and efficiency.
- Implemented secure authentication and authorization using Spring Security and JWT to replace client-side login logic, enabling OAuth 2.0 Google Login and token-based session management; Configured global CORS policy and custom JWT validation filters, ensuring secure communication between backend and frontend.

Backend Developer

12/2025 - 03/2026

Sports Excitement

New York, New York

- Built a Redis-powered rate limiting and session validation cache system that enforced strict limits for API usage and reduced repeated authentication database lookups, preventing abuse while improving application security and stability as the system scales.
- Implemented a Saved-items feature that enables authenticated users to securely bookmark and manage listings with ownership-based access control and comprehensive automated end-to-end integration testing to ensure data integrity and scalability.
- Collaborated with product and design to map user-generated content relationships, defining core entities and their associations, and structuring request/response DTOs to clarify postings workflows and feed rendering.

TECHNICAL SKILLS

Languages: C++, C, JavaScript, TypeScript, Python, HTML, CSS

Technologies: MySQL, Node.js, WebSockets, Firebase, JWT, Spring Boot, Spring Security, NestJS, Docker

AWS: Lambda, DynamoDB, S3, CloudFront, OpenSearch, API Gateway

EDUCATION

University of California San Diego

B.S. in Computer Science & Software Engineering

09/2024 - 12/2026

Diablo Valley College

A.S in Computer Science

08/2022 - 05/2024